Training Language Models to Follow Instructions with Human Feedback

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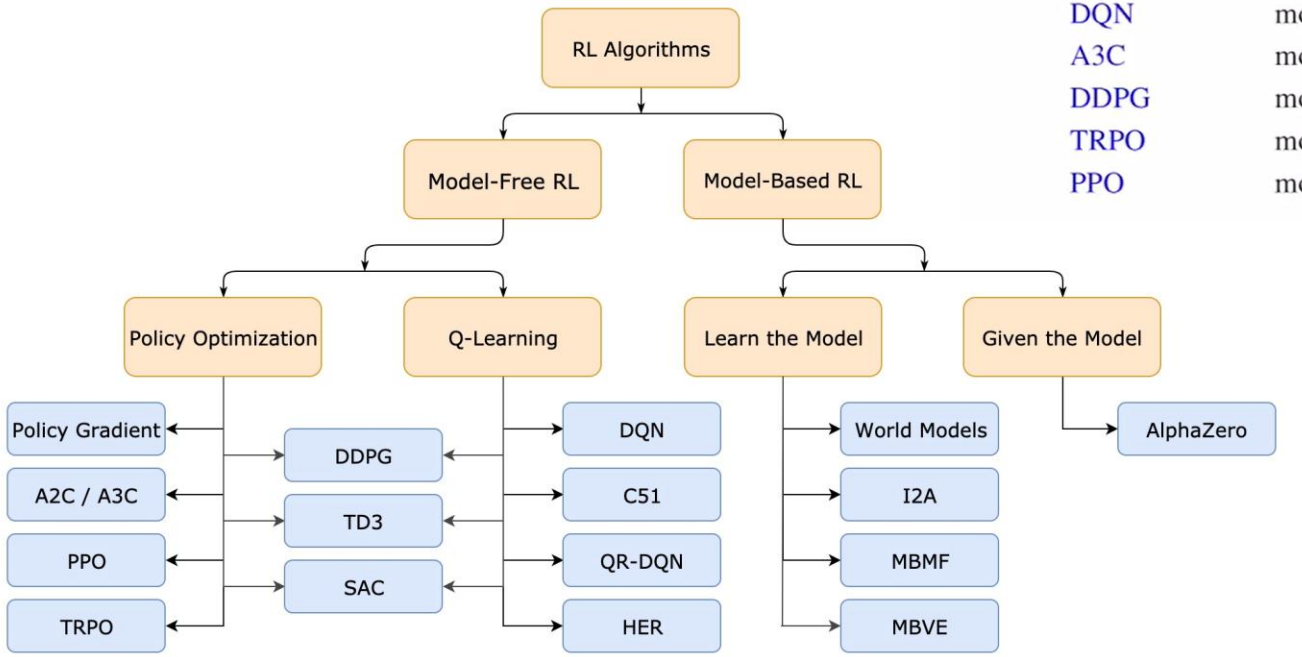
1. Connection of RL Theories to RLHF

So Far about RL Theories…



Comparison of DRL algorithms

Algorithm	Model	State space	Action space	Policy	Objective
SARSA	model-free RL	discrete	discrete	on-policy	Q-value
Q-Learning	model-free RL	discrete	discrete	off-policy	Q-value
DQN	model-free DRL	continuous	discrete	off-policy	Q-value
A3C	model-free DRL	continuous	continuous	on-policy	Advantage
DDPG	model-free DRL	continuous	continuous	off-policy	Q-value
TRPO	model-free DRL	continuous	continuous	on-policy	Advantage
PPO	model-free DRL	continuous	continuous	on-policy	Advantage

Value- Based vs. Policy-Based

Value-Based (e.g., DQN, Q-Learning)

- DQN: discrete **state space** → continuous
  * Q-Value Function: using (s, a) pair
- Optimization: **min Bellman Error**
- **Deterministic**, **Indirect** Policy Extraction (epsilon greedy, argmax)

Policy-Based (e.g., REINFORCE, PPO)

- Policy Gradient: discrete **action space** → continuous
  * Policy Distribution: using trajectory
- Optimization: **max Expected Return**
- **Stochastic**, **Direct** Sampling Policy Extraction (differentiable, gradient ascent)

From TRPO to PPO

$$\mathcal{J}^{\text{TRPO}} = \max_{\theta} \mathbb{E}_{s\sim P_{\text{old}}, a\sim\pi_{\text{old}}}[r(\pi)Q_{\pi_{\text{old}}}] \quad \boxed{s.t.\ \mathbb{E}_{s\sim\rho_{\pi_{\text{old}}}}[D_{\text{KL}}(\pi_{\text{old}}(\cdot|s)||\pi(\cdot|s))] \leq \delta}$$

$$\mathcal{J}^{\text{PPO}} = \mathbb{E}_{s\sim P_{\text{old}}, a\sim\pi_{\text{old}}}[\min(r(\pi)A_{\pi_{\text{old}}}(s,a), \boxed{\text{clip}(r(\pi), 1-\epsilon, 1+\epsilon)A_{\pi_{\text{old}}}(s,a)})]$$

$$\text{where } r(\pi) = \frac{\pi(a|s)}{\pi_{\text{old}}(a|s)}, \quad A_{\pi_{\text{old}}} = Q_{\pi_{\text{old}}}(s_t, a_t) - V_{\pi_{\text{old}}}(s_t) = r(s_t) + \gamma\mathbb{E}_{s_{t+1}\sim P(s_{t+1}|s_t,a_t)}[V_{\text{old}}(s_{t+1})] - V_{\pi_{\text{old}}}(s_t)$$

Transition

Constrained → Unconstrained with Penalty/Clipping.

$$\max_{\theta} J(\theta) \ s.t.\ KL \leq \delta$$

$$\Downarrow$$

$$\max_{\theta} J(\theta) - \beta KL \ or\ Clipping$$

TRPO

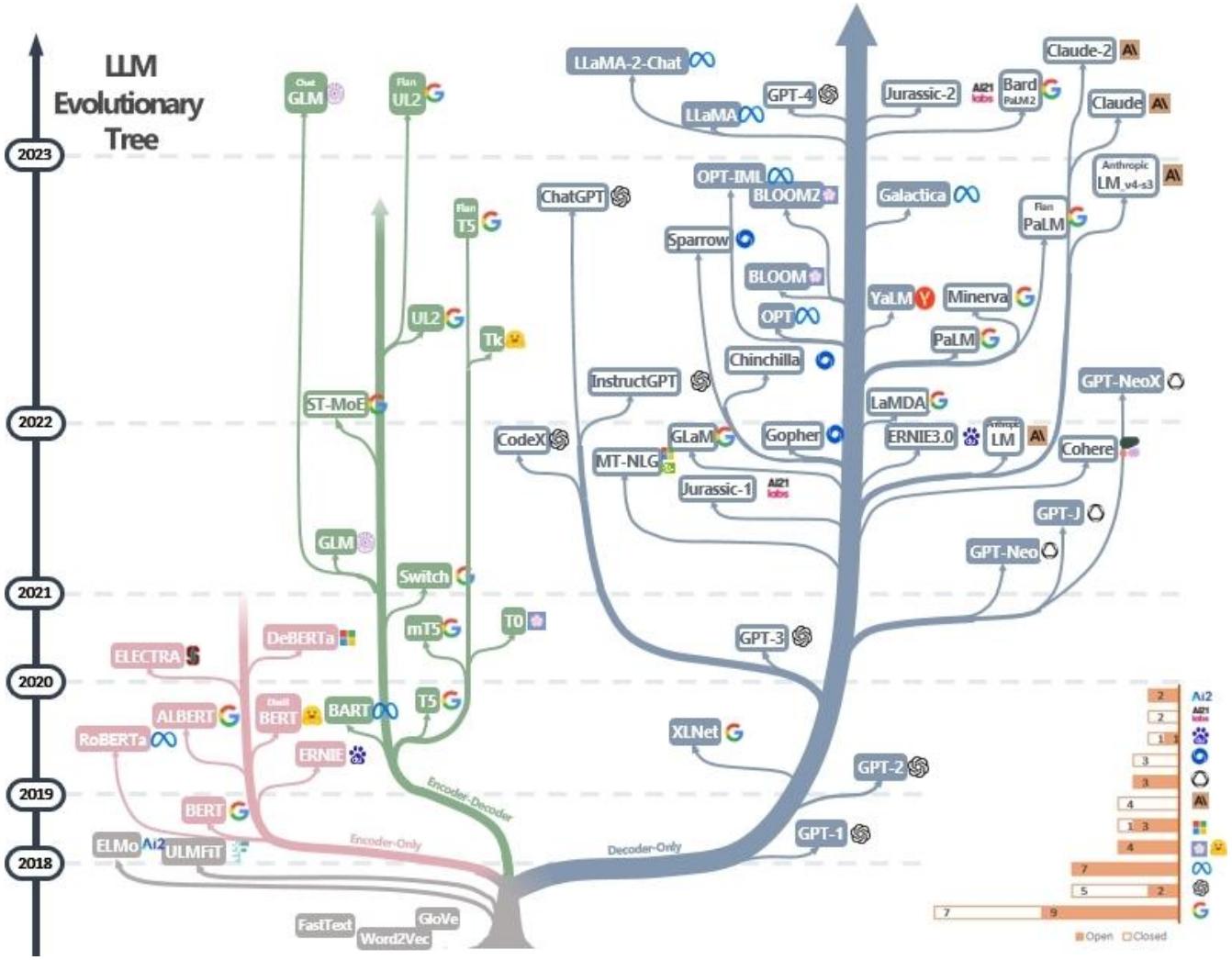
- trust region (KL Divergence) 내에서 parameter update
  → monotonic improvement of the expected return, policy 업데이트의 급격한 변화 제한
- constraint term 처리를 위해 2차 미분 Hessian 계산 필요 (complicate to implement, compute)

PPO

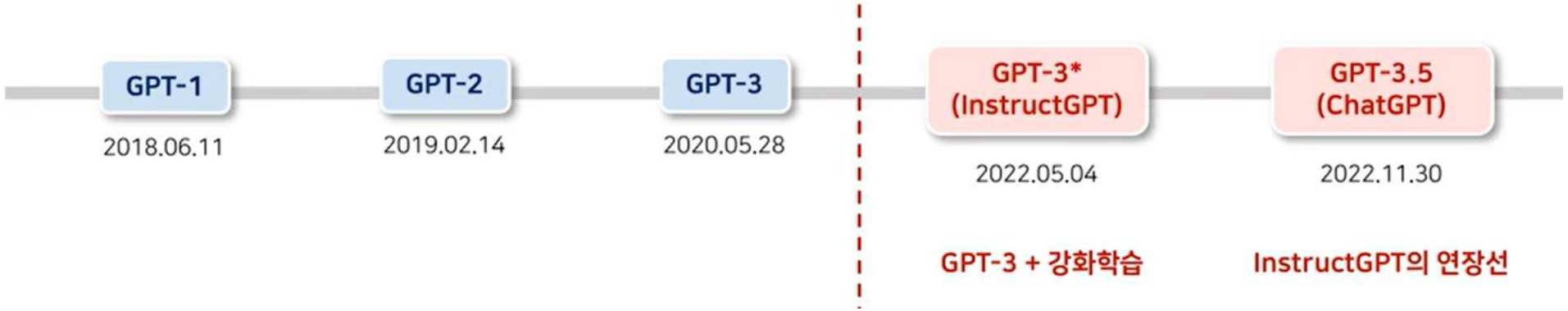
- clipping을 통해 constraint term 처리에 대한 복잡성↓ (1차 미분 Adam, SGD)
- clipping or penalty 처리

1. Connection of RL Theories to RLHF

Recently… In the Era of LLMs !



From GPT3 to InstructGPT



GPT3(2020)까지는 인간의 가치/선호에 대한 고려 X; only next word prediction task

"편향(bias) 없이, 사실에 입각한(no hallucination), 사용자에 도움이 되는(following instruction)"

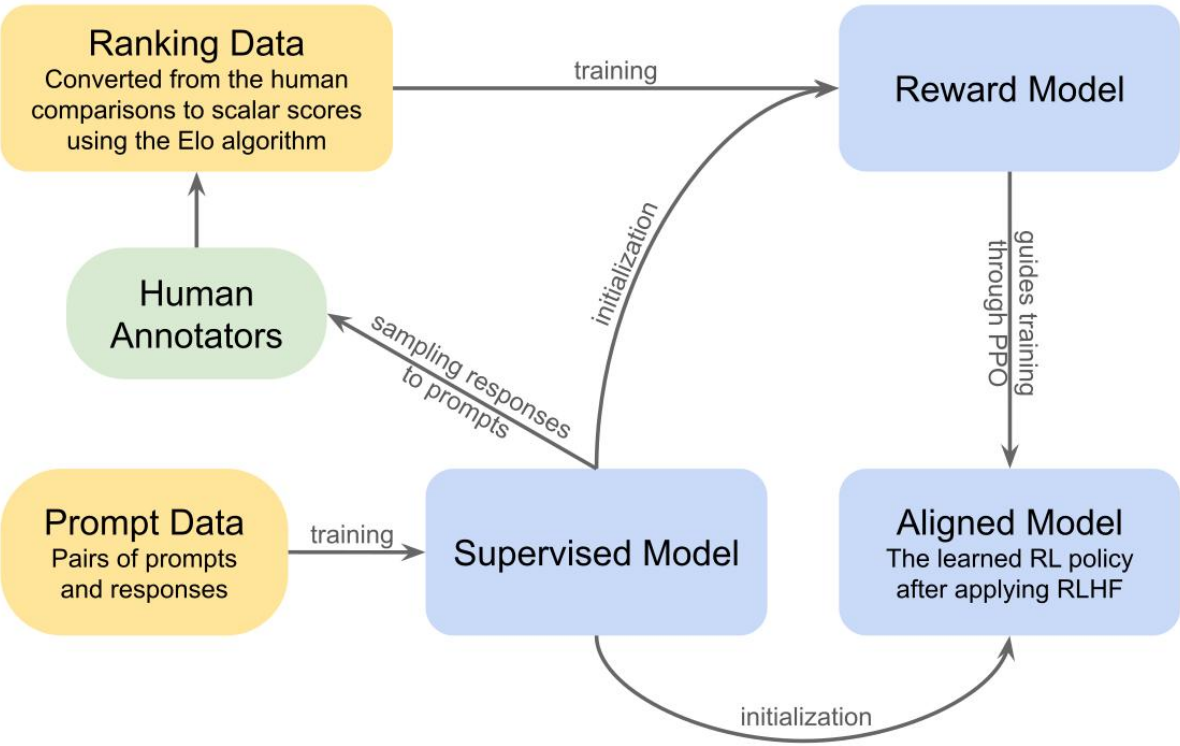
Applying RL to LLM

Agent - LLM(SFT모델)

Policy - next word(token) prediction

Action Space - tokens (= vocab size)

Reward Function - Reward Model



Three Steps for Learning

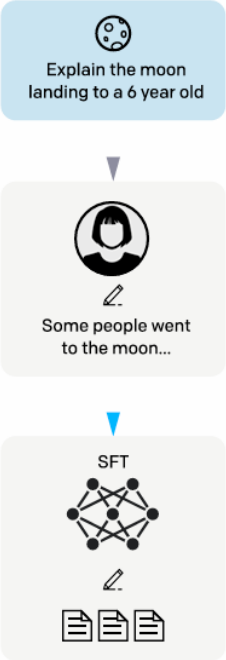
Step 1

Collect demonstration data, and train a supervised policy.

A prompt is sampled from our prompt dataset.

A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3 with supervised learning.



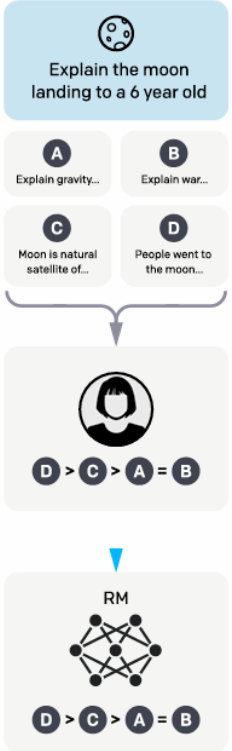
Step 2

Collect comparison data, and train a reward model.

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

This data is used to train our reward model.



Step 3

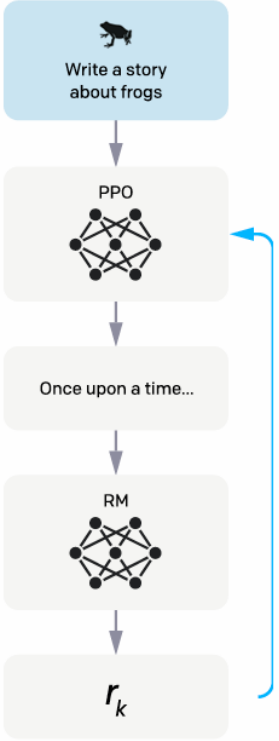
Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.

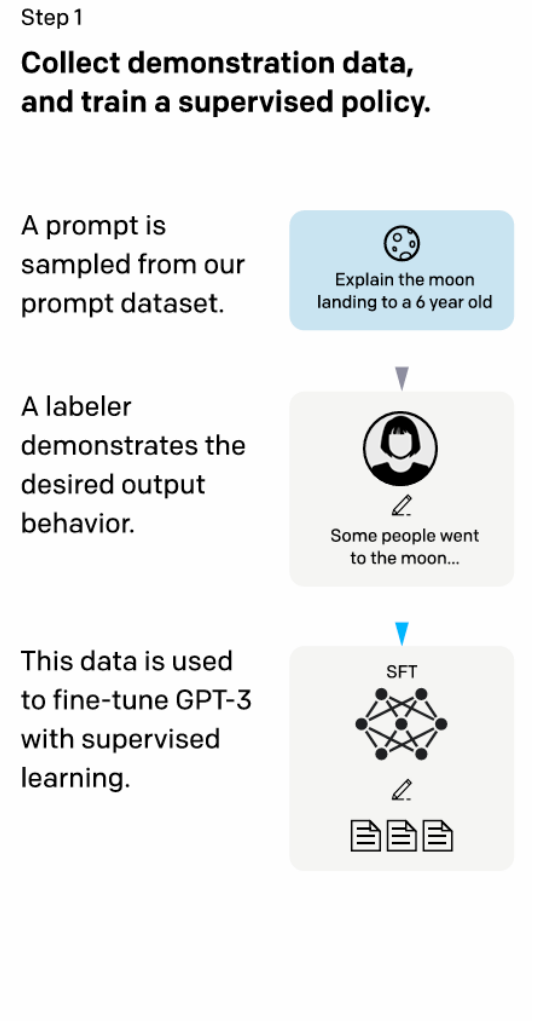
The policy generates an output.

The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.



Step1. Supervised Fine-Tuning



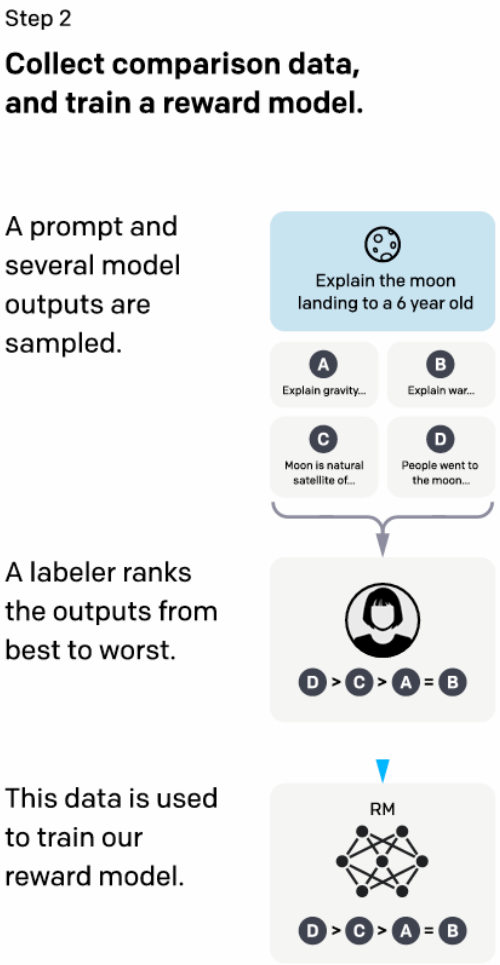
(1) Demonstration 데이터 구축 (prompt : response) 쌍 13K

- Labeler가 직접 response 작성

(2) pre-trained LLM Supervised Fine-Tuning

- 앞서 구축한 데이터셋으로 16 epoch 학습
- Validation: Reward Model을 통한 scoring 결과가 가장 높은 모델 최종 선정

Step2. Reward Model Training



(1) Comparison 데이터 구축 (33K prompt : 4~9 response)

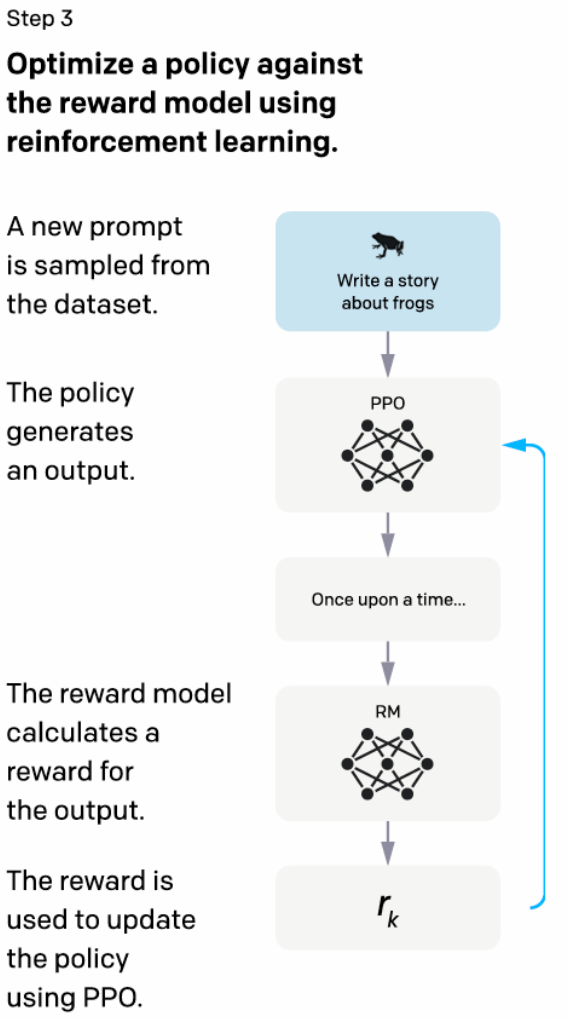
- Labeler가 4~9개의 response 선호도 순서를 labeling

(2) Reward Model 학습

- 앞서 구축한 데이터셋으로 1epoch 학습
- (prompt, response) 입력 / reward(scalar) 출력 모델 학습
- GPT-3 파라미터 초기화 + unembedding layer 변환 학습
- 하나의 prompt 당 두 response 선호도 간의 차이를 기준으로 학습 (1개 prompt 당 k개(4~9) response, reward 차가 클수록 loss 감소)

$$\text{loss}(\theta) = -\frac{1}{\binom{K}{2}} E_{(x,y_w,y_l)\sim D}\left[\log\left(\sigma\left(r_\theta(x,y_w) - r_\theta(x,y_l)\right)\right)\right]$$

Step3. Reinforcement Learning via PPO



(1) Policy Initialization

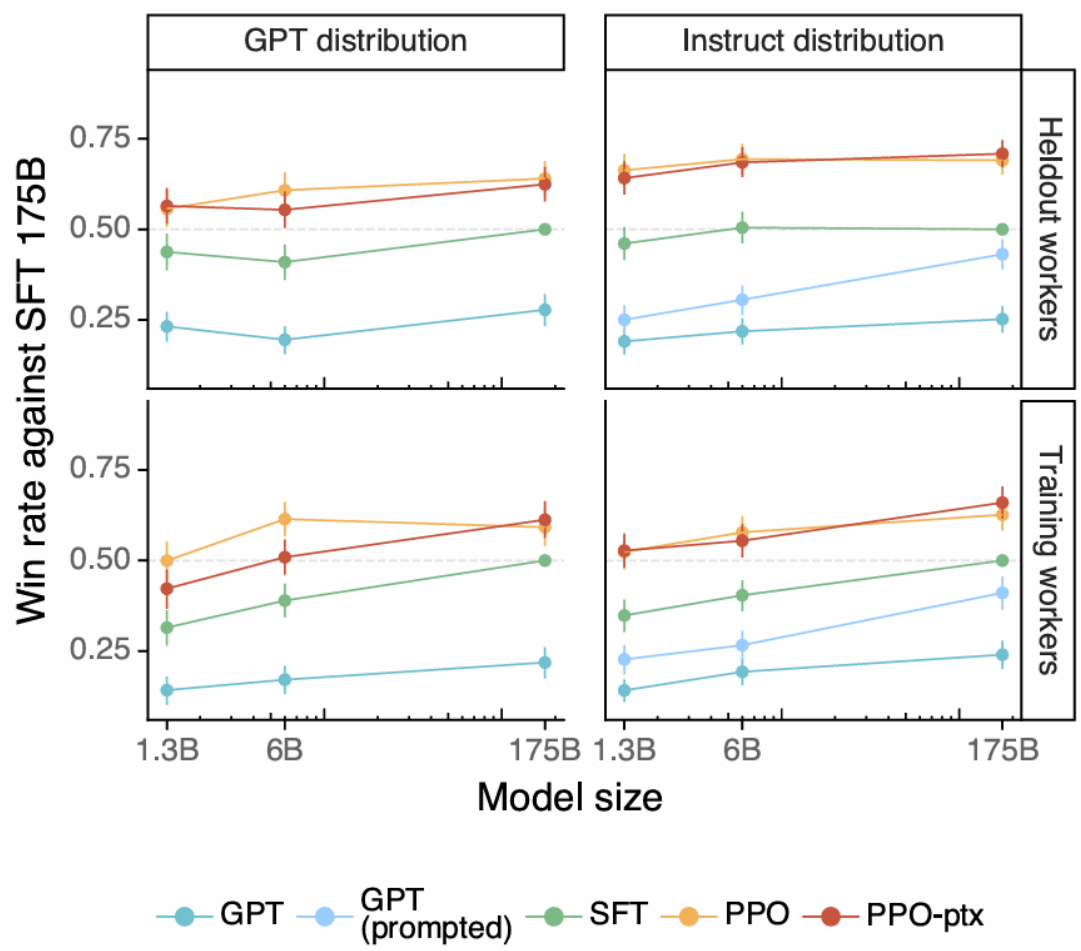
- SFT 방식으로 2epoch 학습된 후의 파라미터 세팅

(2) PPO Fine-Tuning

- 31K의 prompt 데이터셋 구축
- 학습된 Reward Model을 Value Function으로 정의
- 256K 에피소드 학습
- KL penalty 항 추가: 파라미터 변동폭 제한
- cf. pre-training gradients를 PPO gradients와 mix: PPO-ptx term
- cf. 원 논문에서 PPO penalty beta=0.02, clipping epsilon=0.2

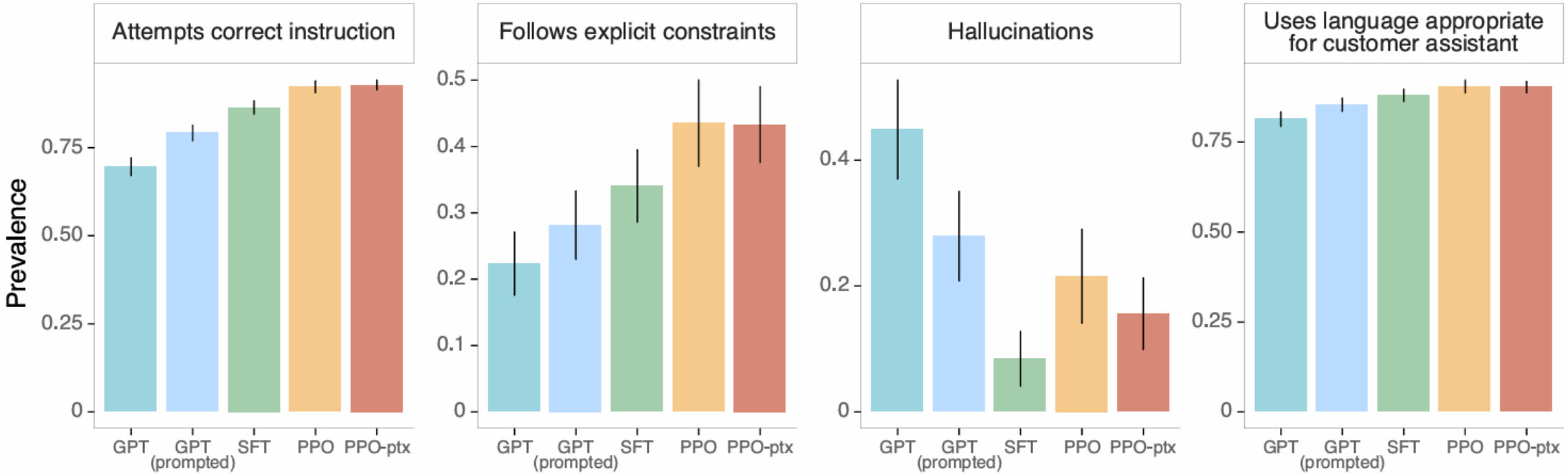
$$\text{objective}(\phi) = E_{(x,y)\sim D_{\pi_\phi^{\mathrm{RL}}}}\left[r_\theta(x,y) - \beta \log\left(\pi_\phi^{\mathrm{RL}}(y \mid x)/\pi^{\mathrm{SFT}}(y \mid x)\right)\right] + \gamma E_{x\sim D_{\mathrm{pretrain}}}\left[\log(\pi_\phi^{\mathrm{RL}}(x))\right]$$

Increasing Human Preference and Generalization



InstructGPT models show promising generalization to instructions outside of the RLHF fine-tuning distribution. We qualitatively probe InstructGPT's capabilities, and find that it is able to follow instructions for summarizing code, answer questions about code, and sometimes follows instructions in different languages, despite these instructions being very rare in the fine-tuning distribution. In contrast, GPT-3 can perform these tasks but requires more careful prompting, and does not usually follow instructions in these domains. This result is exciting because it suggests that our models are able to generalize the notion of "following instructions." They retain some alignment even on tasks for which they get very little direct supervision signal.

Decreasing Hallucination & Improving Following Instructions



Limitations: Regarding Bias

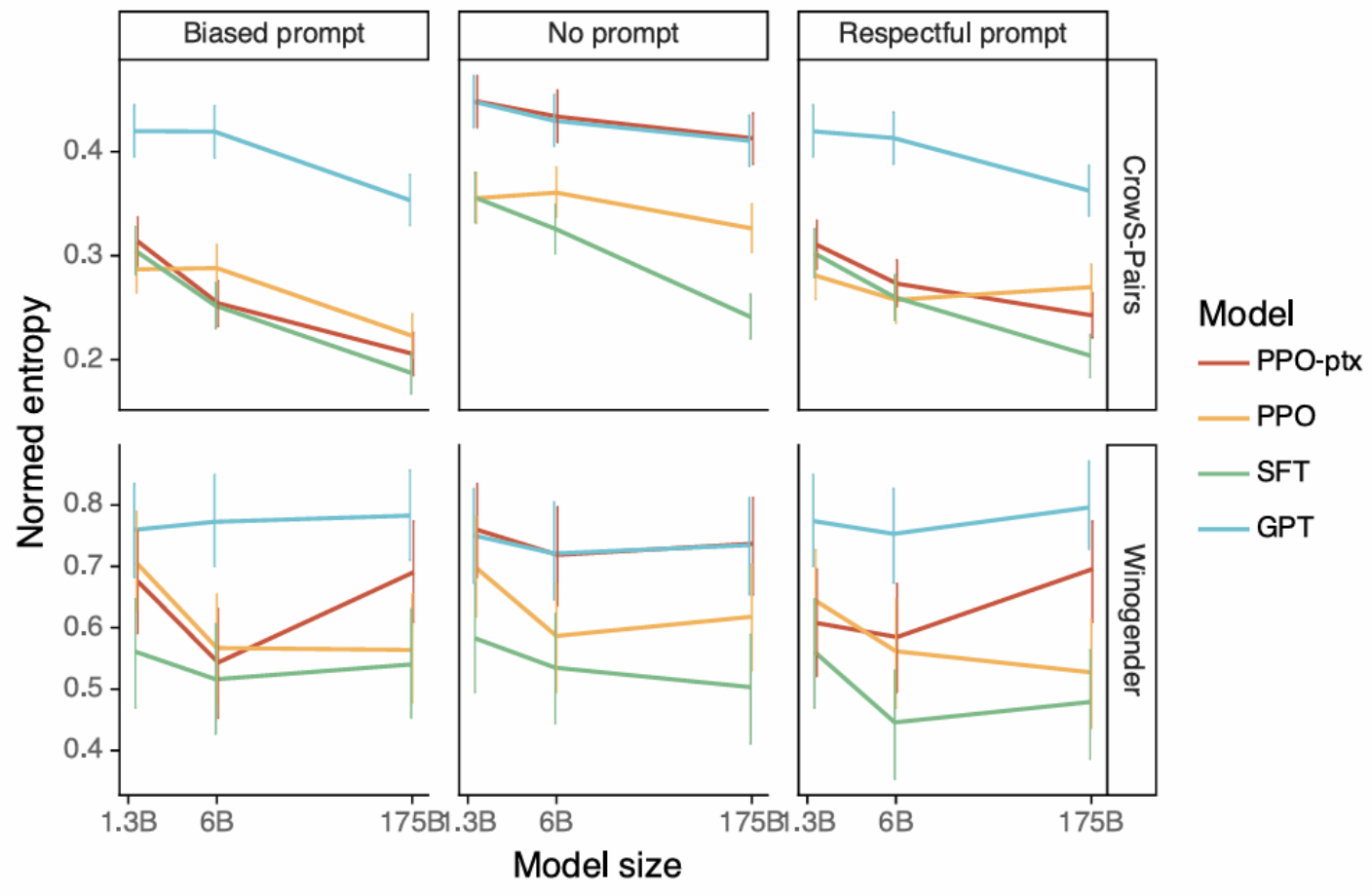
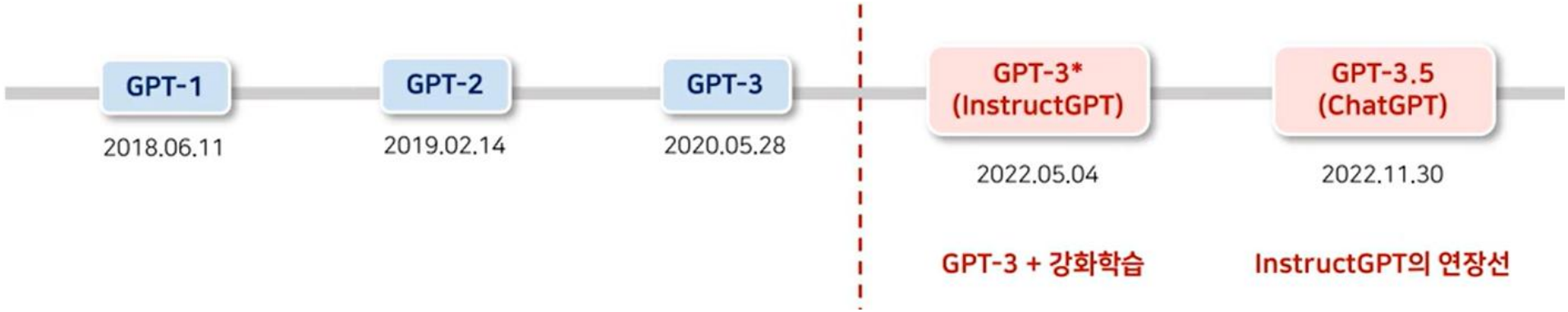


Figure 32: Bias results on Winogender and CrowS-Pairs.

ChatGPT



RLHF는 오늘날의 LLM 챗봇 등장을 가능케 한 Post-Training!

Q&A